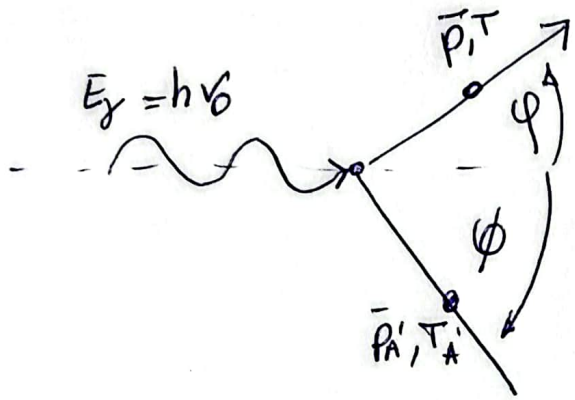
"Compton" para partícula con masa



$$E^2 = p^2c^2 + m^2c^4$$

$$E_{A'}^2 = p_{A'}^2c^2 + m_{A'}^2c^4$$

Conservación $\bar{P}_{total}$

$$\frac{E_\gamma}{c} = p_{A'}c\phi + pc\varphi$$

$$0 = p'_{A'}s\phi - psc\varphi$$

$$\longrightarrow \left(E_\gamma - p_{A'}cc\phi\right)^2 + p_{A'}^2c^2s^2\phi = p^2c^2$$

$$E_\gamma^2 + p_{A'}^2c^2 - 2E_\gamma p_{A'}cc\phi = p^2c^2 \qquad (1)$$

Conservación E_{total}

$$E_\gamma + mc^2 = E_{A'} + E = E_{A'} + \sqrt{p^2c^2 + m^2c^4}$$

$$\left(E_\gamma + mc^2 - E_{A'}\right)^2 - m^2c^4 = p^2c^2 \qquad (2)$$

(1) = (2)

$$E_\gamma^2 + \underbrace{p_{A'}^2c^2}_{E_{A'}^2 - m_{A'}^2c^4} - 2E_\gamma p_{A'}cc\phi = \left(E_\gamma + mc^2\right)^2 + E_{A'}^2 - 2\left(E_\gamma + mc^2\right)E_{A'} - m^2c^4$$

$$-m_{A'}^2c^4 - 2E_\gamma p_{A'}cc\phi = 2E_\gamma mc^2 - 2\left(E_\gamma + mc^2\right)E_{A'}$$

Escaneado con CamScanner

$$\left(2E_\gamma mc^2 + m_{A'}^2c^4 - 2(E_\gamma + mc^2)E_{A'}\right)^2 = 4E_\gamma^2\left(E_{A'}^2 - m_A^2c^4\right)c^2\phi$$

$$\left(E_\gamma mc^2 + \frac{m_{A'}^2c^4}{2} - (E_\gamma + mc^2)E_{A'}\right)^2 = E_\gamma^2\left(E_{A'}^2 - m_A^2c^4\right)c^2\phi$$

Escaneado con CamScanner